

(e) Human-in-the-loop language interface

task Θ

User confirmation
Accept / revise proposed edits
→ structured task Θ
(room, AC, PV, wind, tariff, quotes)

Proposed edits $\Delta\Theta$
field: *from* → *to* (validated against API ranges)
Unsupported items listed, not dropped
Ambiguity → clarification question

Local LLM (on-premise)
Tool-calling under a strict JSON schema
Edits task fields only — never emits results
No data leaves the machine

Decision maker
Natural-language request
"3 units per room,
no grid export"

re-planning loop runs right → left

inspect / revise

(a) Multi-source inputs

Hourly weather

- Reanalysis, 8,784 h (2024)
- Air temp., RH, pressure, irradiance, wind
- Interval semantics normalised

Building & HVAC

- Room geometry, envelope, schedule
- AC catalog: rated capacity, COP
- N rooms × n units per room

Site & equipment

- Roof area → PV capacity bound
- Certified small-wind power curve
- User quotes: PV, wind, storage

Policy & market

- Official 4-period TOU tariff
- Critical peak: Jul–Sep, $T_{\max} \geq 35^\circ\text{C}$
- Grid EF 0.4419 kgCO₂/kWh (GD)

(b) Physics-based hourly simulation

1 Lumped RC heat-moisture room model

$$C \, dT/dt = UA(T_{\text{out}} - T) + Q_{\text{sol}} + Q_{\text{int}} + Q_{\text{vent}} - Q_{\text{AC}}$$

$$m_a \, dw/dt = \dot{m}_w(w_{\text{out}} - w) + G_{\text{lat}} - Q_{\text{lat}}/h_{\text{fg}}$$

$$P_{\text{AC}}(t) = Q_{\text{AC}}(t) / \text{COP}(T_{\text{out}}) \quad \text{sensible + latent, } Q_{\text{AC}} \leq nQ_{\text{rated}}$$

Comfort-gated sizing · 60-min pre-cooling (energy counted)
Shortfall & unmet temperature/RH scored in occupied hours only
Unit sweep $n = 1 \dots n_{\text{max}} \rightarrow n^* = \min\{n : \text{shortfall} = 0\}$ (min. adequate units)

2 PV generation
pvlib (PVWatts) · POA transposition
Cell temperature from air temp. & wind
Capacity bounded by roof area

3 Wind generation
Certified SD6 power curve
Hub-height speed (Hellman law):
 $v_h = v_{\text{ref}}(h/h_{\text{ref}})^\alpha$

4 Hourly energy matching

$$s(t) = \min\{G(t), L(t)\}$$

$$g_{\text{imp}} = L - s \quad e_{\text{sur}} = G - s$$

self-use · grid import · surplus

vs. conventional annual estimate
 $E_{\text{gen}} \times \bar{p}$ (ignores timing & TOU)

(c) Decision analytics

5 Scenario & capacity sweep

S0 grid only S1 PV S2 wind S3 PV + wind

PV capacity $k \in K$ (roof-bounded); S0 kept as baseline

6 TOU-aware lifecycle economics

$$TC = I_0 + \sum_y [\sum_t p(t)g_{\text{imp}}(t) + M_y + R_y] / (1+r)^y - V_{\text{res}}$$

$p(t)$: valley / flat / peak / critical-peak by hour, month and daily $T_{\max}$
Outputs: 10-yr total cost, ΔTC vs. S0 (sign kept), simple payback

7 Carbon accounting

$$\Delta CO_2 = \sum_t s(t) \cdot EF_{\text{grid}}$$

Official provincial factor; carbon price shown as a scenario only

8 Surplus valorisation (add-on paths)

Storage: rule-based dispatch, $\eta = 0.90 \rightarrow$ investment, payback
Grid export: rough revenue from the same surplus

not added to the main ranking

9 Constraint-aware status & recommendation

feasible over budget quote unknown service gap

Recommend $\arg\min TC$ over comparable candidates;
unknown ≠ excluded; service gap never shown as adequate

(d) Decision interface

Progressive computing

Typical-week preview
physics only, < 1 s

Full-year async job
real backend progress

Visual evidence

Energy calendar (month × hour)

Capacity–cost curve (best marked)

10-yr ledger · carbon · storage
Hourly CSV / JSON / printable brief

Accountable output

Every number from computation
Edits invalidate stale results
Saved plans side-by-side
Sources & hashes traceable

hourly flows

tariff $p(t)$, emission factor, quotes

The diagram consists of a light gray rectangular area containing six red circular icons, each with a white letter 'C' and a number (C1 through C6). These icons are arranged in two rows of three. To the right of each icon is a bold title and a descriptive sentence. The contributions are: C1: Hourly coupling (Load, PV and wind matched hour-by-hour under TOU prices, instead of annual energy × average price.); C2: Comfort-gated sizing (AC load and unit count derived from heat–moisture physics with an explicit adequacy rule.); C3: Policy-faithful economics (Official tariff incl. temperature-triggered critical peak, lifecycle costs and provincial carbon factor.); C4: Honest decision semantics (Unknown ≠ excluded; service gaps flagged; grid-only baseline always kept for comparison.); C5: LLM as interface, not calculator (On-premise LLM turns language into validated task edits; deterministic tools compute every number.); C6: Progressive computing (Sub-second typical-week preview, then a full-year job with real progress for interactive use.).

Key contributions	
C1 Hourly coupling	Load, PV and wind matched hour-by-hour under TOU prices, instead of annual energy × average price.
C2 Comfort-gated sizing	AC load and unit count derived from heat–moisture physics with an explicit adequacy rule.
C3 Policy-faithful economics	Official tariff incl. temperature-triggered critical peak, lifecycle costs and provincial carbon factor.
C4 Honest decision semantics	Unknown ≠ excluded; service gaps flagged; grid-only baseline always kept for comparison.
C5 LLM as interface, not calculator	On-premise LLM turns language into validated task edits; deterministic tools compute every number.
C6 Progressive computing	Sub-second typical-week preview, then a full-year job with real progress for interactive use.

C1

Hourly coupling

Load, PV and wind matched hour-by-hour under TOU prices, instead of annual energy $\times$ average price.

C4

Honest decision semantics

Unknown ≠ excluded; service gaps flagged; grid-only baseline always kept for comparison.

C2

Comfort-gated sizing

AC load and unit count derived from heat-moisture physics with an explicit adequacy rule.

C5

LLM as interface, not calculator

On-premise LLM turns language into validated task edits;
deterministic tools compute every number.

C3

Policy-faithful economics

Official tariff incl. temperature-triggered critical peak,
lifecycle costs and provincial carbon factor.

CE

Progressive computing

Sub-second typical-week preview, then a full-year job with real progress for interactive use.